

MITHIL K GOWDA

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Professional Summary

Cybersecurity graduate specializing in DevSecOps, Secure Software Development Lifecycle (Secure SDLC), CI/CD Security, Cloud Security, and Infrastructure Security. Project Intern at ISRO–LEOS with experience developing secure backend applications using Python, Flask, REST APIs, and MySQL. First author of internationally recognized cybersecurity research focused on AI-driven cyber defense, secure software architectures, cryptographic security, and resilient distributed systems. Passionate about integrating security throughout the software development lifecycle and automating secure application delivery. Seeking graduate opportunities in DevSecOps Engineering, Security Engineering, and Cloud Security.

Education

M.S. Ramaiah University of Applied Sciences, Bengaluru

2022–2026

B.Tech. in Information Science and Engineering

CGPA: **8.80 / 10**

Technical Skills

DevSecOps: Secure SDLC, CI/CD Security, DevSecOps Fundamentals, Infrastructure Security, Security Automation, Secure Software Delivery

Cloud & Infrastructure: Google Cloud Platform (GCP), Cloud Security, Cloud Infrastructure Security, Identity and Access Management (IAM), Secure Cloud Deployment

Application Security: Authentication, Authorization, API Security, Secure Coding, Security Architecture, OWASP Top 10

Programming: Python, Java, JavaScript, SQL, HTML, CSS, Bash

Backend Development: Flask, REST APIs, Backend Application Development

Security Tools: Git, GitHub, Postman, Burp Suite, OWASP ZAP, Wireshark, Nmap

Networking: TCP/IP, DNS, HTTP/HTTPS, SSH

Cryptography: AES-256, SHA-256, Zero-Knowledge Proofs (ZKP)

Operating Systems: Linux, Windows

Professional Experience

Indian Space Research Organisation (ISRO) – Laboratory for Electro Optics Systems (LEOS), Bengaluru
2026

Project Intern – Secure Software Development & Data Analytics

- Designed and developed secure backend applications using Python, Flask, REST APIs, and MySQL following Secure SDLC and secure software engineering principles.
- Implemented authentication, authorization, secure API communication, database security controls, and secure backend integration supporting enterprise software delivery.
- Developed scalable backend services emphasizing modular architecture, secure coding practices, cloud deployment readiness, and operational reliability.
- Engineered maintainable backend components supporting secure application development, software quality, enterprise scalability, and resilient software architectures.

DevSecOps Projects & Research

A Multi-Agent AI Cyber Defense Framework Using HoneyBee Method

- First Author and Corresponding Author of an AI-driven cybersecurity framework accepted for presentation at ICIICE 2026 (Dubai, UAE) and accepted as a book chapter in the edited volume *Metaheuristic Optimization for Social Good*.
- Designed a secure distributed architecture integrating intelligent threat detection, adaptive cyber defense, secure software engineering, and resilient application deployment.

- Developed AI-driven security services integrating XGBoost, CNN-LSTM, Deep Q-Network (DQN), adaptive honeypots, reinforcement learning, and automated defensive decision-making.
- Evaluated the framework using CIC-IDS2017 datasets and simulated enterprise environments, achieving 98.24% detection accuracy, 98.51% precision, 98.10% recall, 0.992 ROC-AUC, and a 1.85% false positive rate.

AI-Governed Carbon Credit Exchange with Digital Carbon Passport

- Abstract accepted at the 4th International Conference on Advancing Sustainable Futures (ICASF 2027), Abu Dhabi University, UAE.
- Designed a secure cloud-ready distributed platform integrating authentication, authorization, trust scoring, fraud detection, secure transaction validation, and resilient software architecture.
- Developed the Secure Trade Authorization and Verification Pipeline (STAVP) utilizing AES-256 encryption, SHA-256 hashing, secure authentication, API protection, and secure software engineering principles.

C³T-STAVP Cryptographic Framework (Ongoing Research)

- Designing a secure distributed cybersecurity framework integrating Character Chaffing & Transposition Technology (C³T), Secure Transaction Authorization & Verification Protocol (STAVP), and adaptive trust validation.
- Researching Zero-Knowledge Proofs (ZKP), post-quantum cryptography, semantic obfuscation, privacy-preserving architectures, secure software design, and resilient distributed systems.
- Developing intelligent cryptographic security mechanisms supporting secure software architectures, enterprise applications, and cloud-native security.

Professional Certifications

- Google Cloud Computing – NPTEL
- IBM IT Fundamentals for Cybersecurity Specialization – IBM
- Systems and Usable Security (Elite) – NPTEL
- Ethical Hacker – Cisco Networking Academy
- Introduction to Cyber Security – IGNOU (SWAYAM)
- IBM Front-End Developer Specialization – IBM
- Introduction to Internet of Things (Silver Medal) – NPTEL
- Business Intelligence and Analytics – NPTEL

Achievements

- Selected as Project Intern at the Indian Space Research Organisation (ISRO – LEOS).
- First Author of internationally recognized research focused on AI-driven cybersecurity, intelligent distributed systems, autonomous cyber defense, and secure software architectures.
- Awarded the Reliance Foundation Undergraduate Scholarship (2026).
- Recipient of NPTEL Silver Medal in Introduction to Internet of Things.